Pulse Programmer for Electron Paramagnetic Resonance Spectroscopy

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ABSTRACT: The design of a new pulse programmer is presented. It is based on an application-specific integrated circuit (ASIC) and designed to fulfill the requirements of pulse electron paramagnetic resonance (EPR). The pulse EPR ASIC provides a high time resolution (2 ns) and a high channel density (8 channels per chip). The pulse sequences are controlled by a digital signal processor (DSP). The sequence processing is solved in a general way: the pulse program, written in a high-level Pulse Programming Language (PPL), is interpreted by the DSP, which then calculates a bit image, loads it to the ASICs, and controls the sequence flow. The approach allows any sequence to be easily programmed, and the pulse EPR experiments can be carried out in real time on a repetition rate of up to 2 kHz. The new pulse programmer provides the same or even a higher time resolution and channel density than general-purpose commercial programmers, which are in use in home-built EPR spectrometers. At the same time, the device has a shorter reprogramming time than these programmers and can compete, in particular for 2D experiments, with the existing pulse EPR-oriented commercial systems. In comparison with commercial analogs the software provides the users with a more flexible control over the sequence programming. The capacity of the programmer for work is demonstrated by three-pulse ESEEM and HYSCORE experiments. © 2004 Wiley Periodicals, Inc.
Concepts Magn Reson Part B (Magn Reson Engineering) 21B: 1–10, 2004

KEY WORDS: pulse EPR; pulse programming language; DSP

INTRODUCTION

Since the beginning of pulse electron paramagnetic resonance (EPR), the construction of suitable pulse programmers has been a demanding task. After the progress in pulse EPR methodology (*1*), the number of required pulse channels and the time resolution increase, and the number of acquisition points gets larger with increasing dimensionality. As a consequence, a modern pulse programmer for EPR has to satisfy the following criteria: nanosecond resolution, high density of the pulse channels, short overhead time compared to spin-lattice relaxation times, which can be of the order of hundreds of nanoseconds to hundreds of microseconds at room temperature, and versatile pulse sequence control.

Three kinds of pulse programmers are in use on home-built EPR spectrometers. They are either based on commercial pulse sequence generators or on custom-designed devices or arbitrary waveform generators (AWGs) with digital outputs. The first type of programmers uses stand-alone programmable delay generators to provide the pulse sequence timing and some sort of logic circuits to combine the pulses to sequences (*2*). Custom-designed programmers use fast discrete counters, programmable delay lines (*3*), or a combination of these approaches (*4*, *5*), which

Received 21 July 2003; revised 21 November 2003; accepted 24 November 2003

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Concepts in Magnetic Resonance Part B (Magnetic Resonance Engineering), Vol. 21B(1) 1–10 (2004)

Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/cmr.b.20009

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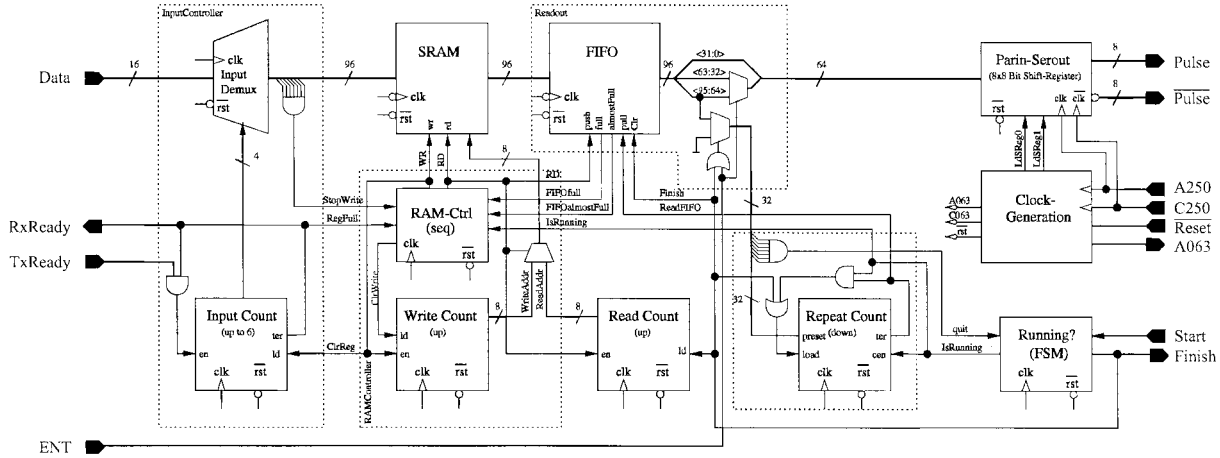


Figure 1 Architecture of the pulse EPR chip. The static random access memory SRAM retains 256 lines of 96-bit words. The ASIC accepts data through a 16-bit input controlled by a two-line RxReady/TxReady asynchronous handshake protocol. Input demultiplexor Demux, memory controller RAM-Ctl, and two up-counters (Input and Write) manage the data loading. A first-in first-out FIFO memory is used for buffering the data stream between the SRAM and the parallel-input serial-output shift register Parin-Serout, which is not sending and receiving at exactly the same rate. The register accepts 64-bit words at 62.5 MCycles/s and converts these words to a bit pattern (8 bit deep, fired sequentially at 250 MCycles/s) on 8 pulse channels (Pulse). The down counter (Repeat) uses the 32 high-order bits to control the number of repetitions of the pattern. A250/C250 is the 250 MHz differential main clock; A063/C063 is the 62.5 MHz generic clock; ENT is the control signal to enable the test mode in which the sequence is repeated continuously.

also include sophisticated logic arrays and sometimes on-board microprocessors. Recently, the third type of the programmers became popular, an AWG1000-DOUTS board (Chase Scientific Co., Langley, WA; AWG1000/1200 User Manual. http://chase2000.com/awg1000/awg1000.htm.) has been implemented by several EPR groups (O. Poluektov, personal communication).

Most of these programmers do not satisfy at least one of the criteria mentioned above. Typically, the overhead time, which is the time needed to start a new sequence that is different from currently played, is too long. It should be mentioned that the overhead time can include the time to reload the data pattern or not, depending on a hardware design. The overhead time for the above-mentioned general purpose programmers lies in the 30–100-ms range and is <30 ms for a device with programmable delay lines operating in "stepping on" mode (*3*). Note that in the latter case the pulse pattern has to be loaded only once for entire experiment. With the programmer of the Novosibirsk group (*5*) and the commercial PatternJet from Bruker, the sweep in one-dimensional (1D) experiments is performed on hardware and does not require a reloading of the pulse pattern, resulting in short rearm times. Recently, the reprogramming time of $\sim$2 ms for 1D or 2D experiments has been achieved on the AWG1000-DOUTS board (A. Astashkin, private communication).

To satisfy all the criteria we have chosen a solution based on recent developments in the field of high-speed electronics and microprocessors. We designed an application-specific integration circuit (ASIC), which fulfills the requirements for high time resolution (2 ns) and high channel density (8 channel per chip, 2 chips on the board). Furthermore, we implemented a digital signal processor (DSP) for the pulse sequence control and developed a compiler for the programs written in Pulse Programming Language (PPL) (*2*). The new pulse programmer was tested on a recently updated X-band pulse EPR spectrometer that is similar to the one described in Wacker (*6*).

MATERIALS AND METHODS

Pulse EPR ASIC

The ASIC was designed and tested by the Integrated Systems Laboratory at ETH, Zurich. The chip architecture is shown in Fig. 1. It has an internal static

random access memory (SRAM) with 256 96-bit words. Each word represents 64 bits for the bit pattern and 32 bits for the pattern repeat count. Hence the 64-bit pattern can be repeated (2^{32}–2) times. The largest possible number, (2^{32}–1), of the repeat count is reserved as termination signal for sequence processing. The loading of the patterns is done in an asynchronous way as is shown in Fig. 2(a). The ASIC notifies that it is ready to receive data by an RxReady signal. To signalize the validity of the data, the provider has to raise the TxReady signal. The rising of TxReady must happen when RxReady is high and falling of TxReady is only allowed when RxReady is low. The chip accepts data at the rising edge of an internal generic clock A063 (62.5 MHz), when both the TxReady and the RxReady signals are high, which cases RxReady to fall. The interface is able to catch data at a rate of up to 60 MCycles/s, but a slower rate is also allowed; the RxReady signal will not go high again while TxReady is high.

The conveyer operation (processing a given memory line and simultaneous overwriting the previous one) is permitted and allows further reduction of the re-arm time. The generic clock is derived from the main clock A250 (250 MHz) by dividing its frequency by 4. After receiving a Start signal (with length of at least one cycle of the A063 clock) the memory content is transmitted word by word at 62.5 MSamples/s through an FIFO buffer to the output Parin-Serout shift register.

The timing diagram of the sequence launch is shown in Fig. 2(b). The start of the pulse sequence is delayed by 13 cycles of the main clock with respect to the time when the ASIC accepts the Start signal. The parallel-in/serial-out shift register transforms the incoming 64-bit words to sequences of 8 bits on 8 independent output channels, as is shown in Table 1. The register works at a clock frequency of 250 MHz and produces pulse patterns with 2-ns resolution, using both raise and fall edges of the clock. Figure 3 indicates the output timing for the individual bits of the output word. The used bit naming convention is shown in Table 1. When execution of the sequence is terminating, the Finish signal is raised by the ASIC, as is shown in Fig. 2(c). A previously executed pulse sequence can be restarted by rising Start again immediately after Finish.

The main clock and the output pulse signals are of LVDS type. The input interface consists of a 16-bit wide data bus, handshake lines, and sequence control signals. All these signals are of CMOS type. Austria Micro Systems process technology was used (0.6 μm, triple metal, double poly, CMOS n-well, "CUP"). One chip contains 3202 standard cells and two memory blocks. The die area is 14.6 mm^2. There are three separated power domains (+5 V): the core logic, the CMOS input and output buffers, and the LVDS receivers and transmitters.

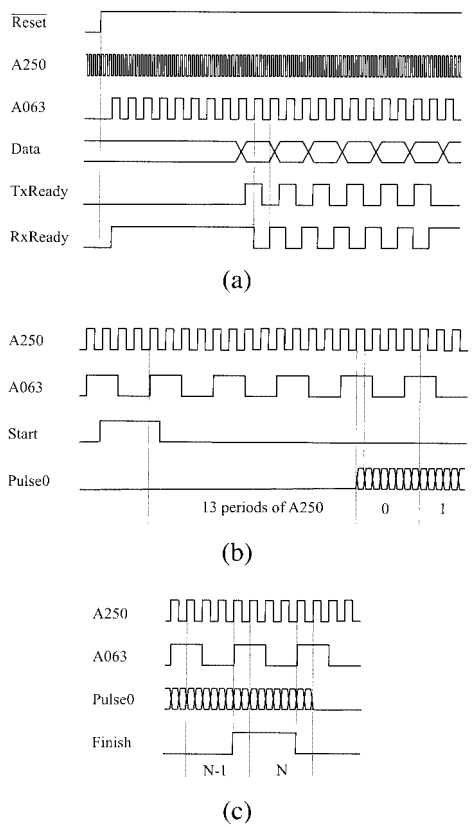


Figure 2 Handshake protocol and timing diagrams. (a) Data loading. The ASIC shows that it is ready to accept data, by raising the RxReady signal. A valid data word is provided onto Data pins. After the data has settled, the TxReady signal is raised high. The chip signals the acceptance of the data on the next rising edge of the A063 by pulling RxReady low. The TxReady signal is pulled low. The Reset signal is shown for completeness and is required in cases of an initialization and of an error. (b) Run of the pulse sequence. The sequence is executed by raising the Start signal. The high time of the signal should last one cycle of the A063 or longer. There is a constant delay of 13 cycles of the A250 between the acceptance of Start and the beginning of the pulse sequence. The sequence is shown for one output channel Pulse0. Numbers beneath Pulse0 are processed memory lines. (c) Finishing. When the execution of the sequence is terminating, a Finish signal is raised by the ASIC. The Finish lasts for one period of the A063. The previously loaded sequence can be launched again after Finish following to diagram (b).

Table 1 Bit Naming Convention for the Pulse Pattern

Memory Line	Repeat Count (32 bits)	Channel Data (64 bits)[a]			
0	$R_{0,31} \ldots R_{0,0}$	$B_{0,7,0}$	$B_{0,6,0}$		$B_{0,0,0}$
		$B_{0,7,1}$	$B_{0,6,1}$	$\cdots$	$B_{0,0,1}$
		$\cdots$	$\cdots$		$\cdots$
		$B_{0,7,7}$	$B_{0,6,7}$		$B_{0,0,7}$
1	$R_{1,31} \ldots R_{1,0}$	$B_{1,7,0}$	$B_{1,6,0}$		$B_{1,0,0}$
		$B_{1,7,1}$	$B_{1,6,1}$	$\cdots$	$B_{1,0,1}$
		$\cdots$	$\cdots$		$\cdots$
		$B_{1,7,7}$	$B_{1,6,7}$		$B_{1,0,7}$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
n	$R_{n,31} \ldots R_{n,0}$	$B_{n,7,0}$	$B_{n,6,0}$		$B_{n,0,0}$
		$B_{n,7,1}$	$B_{n,6,1}$	$\cdots$	$B_{n,0,1}$
		$\cdots$	$\cdots$		$\cdots$
		$B_{n,7,7}$	$B_{n,6,7}$		$B_{n,0,7}$

[a] $B_{i,j,k}$ represents a bit on the ith bit pattern line for pulse channel j, which is in position k in the output timing shown in Fig. 3. The repetition of the pattern is indicated by the repeat count $R_{i,m}$.

The implemented solutions consisting of a compact memory (256 word), a 32-bit repeat count (one memory line /delay interval), and a simple IO interface, allow for an easy programming of the sparse sequences used in EPR. This greatly simplifies fabrication, testing, and implementation of the chip. In a first series 8 of 10 chips passed successfully the test on the HP83000 ASIC verification system (Hewlett-Packard).

System Layout

The configuration of the complete system used for the testing experiments is shown in Fig. 4. The single-board computer (controller) SBC62 (2; Innovative Integration) is implemented to control the pulse sequences. The SBC62 is equipped with a TMS320C6201 processor (Texas Instrument, 180 MHz, 1600 MIPS). It provides the interfaces to the host computer (*1*) and to the pulse forming board (3; PFB). The controller-to-PFB interface consists of a 16-bit wide data bus and 8 control lines. Five control lines, TxReady (ready-to-transmit), RxReady (ready-to-receive), Reset, Start, and Finish are required for a single chip as is shown in Fig. 1. On the controller side a 32-bit IO data register is used to drive the interface lines. It is configured as 3×8 output lines and 8 input lines. The Start line can be driven by an output of the IO register (software trigger), by an AD9850 synthesizer of the SBC62 board, or, optionally, by an external source. The Finish control line connects the external interrupt line of the controller to the PFB flip-flop, which is activated by the ASIC Finish signals. Two control lines are used to select on-board ASICs, and the remaining control line serves to clear the Finish flip-flop. Two on-chip timers of the SBC62 are used to control the sequence repetition rate (the accuracy corresponds to two processor

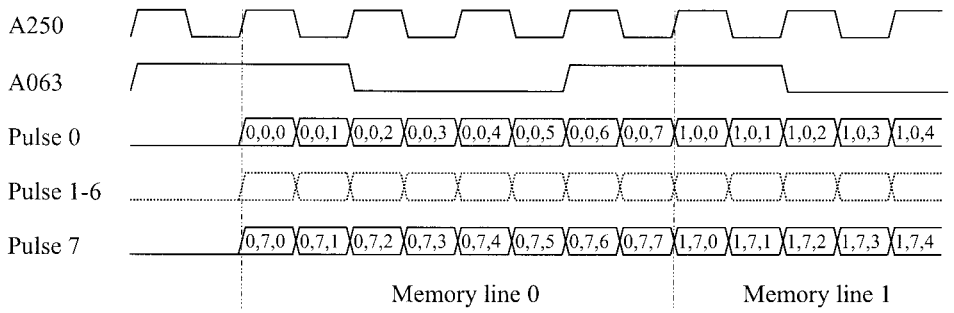


Figure 3 Pulse bit output sequence and timing with respect to the A063 and A250 clocks. The used bit naming convention is shown in Table 1.

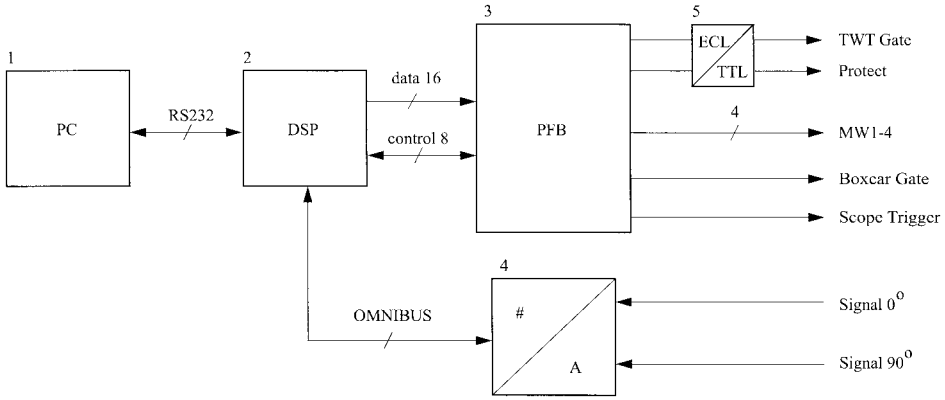


Figure 4 Hardware setup and pulse channel configuration used for the tests. (*1*) host PC; (*2*) SBC62 single board computer; (*3*) pulse forming board; (*4*) ADC plug-in board for SBC62; and (*5*) ECL-to-TTL converter. The ADC is triggered through OMNIBUS under program control.

clocks) and the time to trigger an analog-to-digit conversion on the ADC board (*4*).

The PFB (*3*) is equipped with two chips leading to 16 outputs, a (250 ± 100 ppm) MHz clock oscillator (EH01-531, Connor-Winfield Corp.), an interface circuit and 50-Ω ECL drivers. An external ECL to TTL converter (*5*) matches the signal logic with the logic of the recipient. In our case the gate signal for the pulsed traveling wave tube (TWT) amplifier and the driving pulse for the receiver protection switch have to be TTL type.

The system is equipped with analog-to-digital converters (*4*) to acquire the signals coming from the quadrature receiver of the pulse EPR spectrometer. For this purpose, an A4D4 plug-in board (Innovative Integration) for the SBC62 is implemented. The board has four 16-bit ADCs, which work at a conversion rate of up to 200 kHz. The triggering of the ADCs is synchronous in pairs; the reading from two ADCs is performed in a single step through a 32-bit peripheral bus of the controller (OMNIBUS). To digitize the transient signals, an electron spin echo in the case at hand, two gated integrators (SR250, Stanford Research Systems) were implemented in front of the A4D4 (not shown). These integrators were used in a sample-and-hold mode and signal averaging was performed digitally.

Software

An important issue is the software support of the programmer. To reduce the time to implement a new pulse experiment, the software should provide an effective and intuitive user interface. The most general solution is to use a high-level language for pulse sequence programming. To program pulse sequences a slightly modified version of the PPL used at the Weizmann Institute of Science (*2*) and at ETH (*7*) has been implemented. The new version is based on the grammar mentioned above. It defines the types of data, commands, pulse operations, and arithmetic and Boolean operations as well as the actions that have to be performed in accordance with program statements [see Shane et al. (*2*) for further details]. In the new version the actions are to create an intermediate assembler-like representation of the pulse program that can be interpreted by the controller. The compiler, which translates the PPL program to this representation, was created using a Bison parser generator (*8*). The compiled code is sent to the controller. Then the code is interpreted in terms of an events list ordered according to the pulse sequence. Finally, the bit pattern is created in accordance with the memory organization of the ASIC as is shown in Table 1. The controller program also performs a check of the pulse behavior. The critical pulse EPR-specific signals are the gate signal for the TWT amplifier and a protect pulse, which has to follow the TWT gate to avoid destruction or an overloading of the receiver amplifiers. In addition, the duty cycle of the sequence and the maximal pulse length have to be validated in accordance with the technical specifications of the TWT amplifier. An example for a PPL program is shown in Fig. 5a. More examples can be found in Shane et al. (*2*).

The DSP program is written in C using Code Composer Studio (TI). The program consists of a simple stack-based interpreter, a host, a PFB and user interfaces, and a 3D sweep engine. The DSP-PFB interface uses 32-bit IO register and emulates the data

(a)

```
time t1 = 128 ns
time t2 = 128 ns
time tau =470ns
const time t90 = 16 ns
const time t180 =16 ns

program HYSCORE
mwpulse t90, [3, 3, 3, 3]
wait tau
mwpulse t90, [3, 3, 3, 3]
wait t1
mwpulse t180, [1, 1, 2, 2]
wait t2
mwpulse t90, [3, 4, 3, 4]
trigger 1
wait tau
detect re=[1,-1,1,-1]
```

(b)

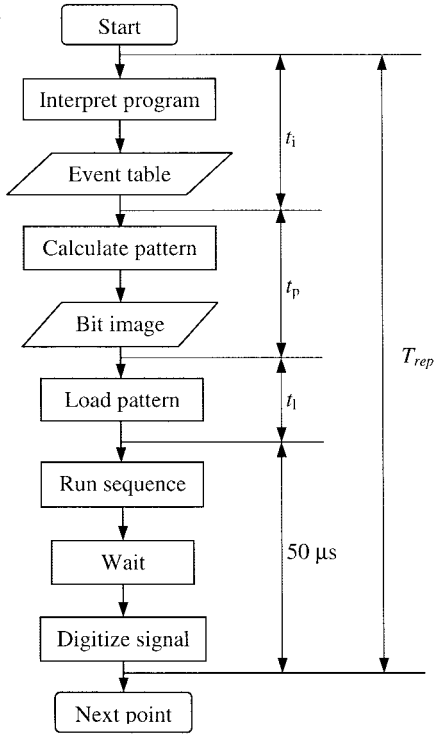


Figure 5 Pulse sequence processing. (a) Program example. (b) Program flow for a single point measurement. The most inner loop of the sequence processing is shown. The main time intervals are labeled on the right side.

loading protocol of the ASIC. The writing is performed in three steps: 1) the program checks the RxReady signal on a corresponding pin of the IO register; 2) if RxReady is true, a 16-bit word is written to the register and the TxReady bit is set true; and 3) finally, the TxReady bit is set false. This read-write-write cycle takes about 0.8 μs. The sequence bit image is loaded to the memory using packages of six 16-bit words. A typical memory usage for pulse EPR experiments lies in the range of 20–40 memory lines, so that up to 240 words are transmitted. In the case of a writing error, RxReady is false on step 1 above, and the full sequence is reloaded. This way of error recovering is acceptable, because the up-loading time is anyway short, about 190 μs for 40 memory lines, and the error level on the test bench was better than about 1 error per 12,000 writing cycles.

The controller program includes a sweep engine, two loops to sweep the time intervals (*x, y*), a sweep accumulation loop (scans), and a point accumulation loop (shots). To control the repetition rate and the ADCs reading time, the on-chip timers of the controller were used via interrupt procedure. In addition, to control the end of the pulse sequence in real time, the program can process an interrupt request driven by the Finish signal of the PFB. The program flow diagram (most inner loop, single point) is shown in Fig. 5(b).

RESULTS

The timing parameters of the programmer were checked with a WavePro 960 (LeCroy) digital oscilloscope with 2-GHz analog bandwidth and 16-GS/s sampling rate. The ASIC is a linear device and the instability of the clock oscillator is translated to the fluctuation of the pulse sequence intervals. These fluctuations are very small and could not be observable under our test. However, because both raising and falling edges of the clock are used to produce the pulse sequence, it was necessary to tune the duty cycle of the main clock as close to 2 as possible. The eight outputs of a single chip are sufficient to perform pulse EPR experiments with four mw channels and to provide the triggers for the boxcar and the ADC and to control the TWT and the receiver protection gates. However, some advanced experiments require more mw channels or radio-frequency (rf) channels are

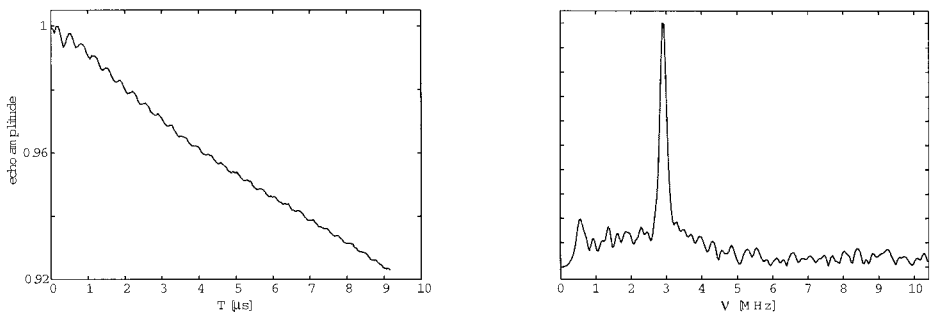


Figure 6 Three-pulse ESEEM experiment on γ-irradiated quartz glass. (a) Echo amplitude in the time domain. (b) Fourier transform of trace (a). Base line correction, apodization with a Gaussian window and zero filling were performed prior FFT. Four thousand traces were accumulated (10^3 sweeps × 4 phases) with a repetition rate of 1.25 kHz. Two hundred points per trace were measured. The extremely weak modulation (<0.5%) is due to matrix ^{29}Si nuclei. Other parameters are as follows: ν_{mw} = 9.62 GHz, B_0 = 343.7 mT; dwell time, 48 ns; single shot measurement.

needed (*1*). Consequently, two or more chips have to be used and chip-to-chip jitter becomes important. In the case of two chips the jitter was found to be 16 ns, which is one period of the generic 62.5-MHz clock. This is due to the simple triggering circuits implemented on the test PFB. Such a jitter is too large for the mw channels but still acceptable for rf channels, in particular in the case of polarization transfer double resonance experiments (*1*).

To test the programmer two standard pulse EPR experiments were carried out on γ-irradiated quartz glass (Herasil) at room temperature and at an mw frequency in the X-band range. The measurements were performed in a silent mode, i.e., the accumulated data was not transferred to the host computer during acquisition. A single-shot mode was used to test the programmer in the complicated situation, where the sequences have to be recalculated and reloaded for each data point. It is worth to mention that once the sequence is loaded it can be played an unlimited number of times (e.g., for multiple shots measurement). An additional comment about single-shot measurements should be made. In pulse EPR spectroscopy one of the most frequently used approach for signal averaging is to repeat the sequence at each point n times, where n is the number of shots. The effective sampling rate corresponds then to the repetition rate per number of accumulations, i.e., it lies in the Hertz range, where electronic noise and spurious signals are profound. Alternatively, averaging can also be done by sweeping the time interval with n = 1 and repeating this procedure m times, where m is the number of scans. This approach suffers less from low-frequency noise and is recommended for experimental setups that are not optimized with respect to low-frequency instabilities (i.e., temporary, experiment specific, constructions, etc.).

In a first experiment, the modulation of a three-pulse electron spin echo (*1*) caused by ^{29}Si nuclei (natural abundance, 4.7%) was measured. The delay time τ between the first and the second pulse was fixed and the delay time T between the second and the third pulse was swept point-by-point. To eliminate unwanted signals, a four-step phase cycle was used, in which the phase of the first two pulses and the phase of the third pulse are alternated by 180° (*9*). In this experiment six pulse channels were used, two of them to drive the mw pulse channels and four for the remaining channels (see Fig. 4). The modulation depth in this sample is only ~0.5% of the echo amplitude (*10*) and as a consequence extensive data accumulation was required. The signal-to-noise ratio for the echo at the initial value of the delay time T was ~40. The echo modulation becomes visible after ~10 accumulations. The phase cycle was applied for each value of T and the incrementation of T was repeated to increase S/N (x-cycle-scan). At each point the PPL program was interpreted, the event table was built, and the bit pattern was calculated and downloaded to the chips, as is shown in Fig. 5(b). The modulation of the three-pulse echo is shown in Fig. 6(a). Four thousand traces were accumulated, 1000 sweeps × 4 phases, at a repetition rate of 1.25 kHz, and 200 points per trace were measured. The total measuring time was 640.047 s. The overhead time of 47 ms is caused by a few loading errors, which were corrected by the

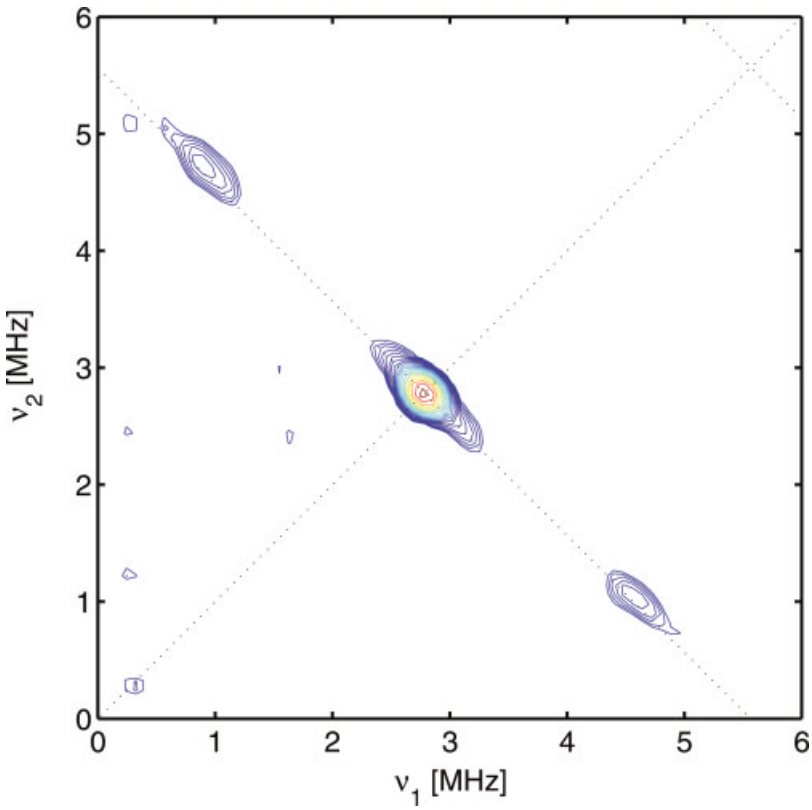


Figure 7 HYSCORE spectrum of γ-irradiated quartz glass. (325 × 4) sets of (128 × 128) data blocks were accumulated at 1 kHz. The total measuring time was about 6 h. The correlation peaks at (1, 4.7) and (4.7, 1) MHz are due to ^{29}Si nuclei close to the E′ center. The conditions are the same as shown in Fig. 4, except for ν_{mw} = 9.20 GHz and B_0 = 328.8 mT. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

program (the data were reloaded again). It was found that this is a typical value for the setup under test. The experiment demonstrates that at a repetition rate of 1.25 kHz the reloading of the pulse programmer chips can be done between the trigger events. The peak at about 2.9 MHz in the spectrum obtained by Fourier transformation [Fig.6(b), S/N ≈ 20] is due to the very shallow modulation caused by matrix ^{29}Si nuclei [Fig.6(a)]. The absence of artifacts in the spectrum indicates that the programmer (hardware and software) provides a correct processing of the experiment.

Next we carried out a Hyperfine Sublevel Correlation (HYSCORE) experiment with the pulse sequence π/2-τ-π/2-t_1-π-t_2-π/2-τ-echo (*11*), a very popular 2D technique, which requires extensive manipulation of the pulse sequence. Two time intervals are swept, and a four- or eight-phase cycle is required. As a consequence, the reprogramming time becomes important.

The PPL program for HYSCORE is shown in Fig. 5(a). The arguments of the *mwpulse* command are a time and a phase, which in our case corresponds to the number of the mw channel (each mw channel is tuned to the proper phase). For the *detect* command the arguments correspond to the sign of the signal. The phase cycle is the index in the arrays. HYSCORE was performed in a way similar to the stimulated echo experiments: the t_1 time interval was swept, the phase cycling was applied at each point, the sweeps of time t_1 were repeated for signal averaging, and then time t_2 was swept (x-cycle-scan-y). (325 × 4) sets of 128 × 128 data blocks were accumulated at a repetition rate of 1 kHz. The correlation peaks at (1, 4.7) and (4.7, 1) MHz in Fig. 7 are due to ^{29}Si nuclei close to the E′ center.

The total measuring time was about 6 h (1 × 325 × 4 × 128 × 128 ms), with only a few seconds of overhead due to loading errors. This compares

Table 2 Time Usage on Different Stages of the Sequence Processing as is Shown in Fig. 5(b)

Sequence	Used Memory (lines)	t_i (μs)	t_p (μs)	t_1 (μs)	T_{rep} (μs)
ESEEM	29	138	332	137	657
HYSCORE	34	172	418	164	804

with the overhead time for the PatternJet programmer of the E580 console (Bruker), which in the case at hands (x-cycle-scan-y) is $300 \times 4 \times 325 \times 128$ ms $\cong$ 14 h due to ~300 ms reprogramming time for each point in the second dimension and each step in phase cycle. Of course the PatternJet is usually not used for scans with a single shot per phase. It is worth to mention that in the case of GPIB-based programmers the situation is even worse; the reprogramming time in both dimensions is ~100 ms. The given example demonstrates the technical advantage of the implemented solution (ASIC design and PPL-DSP-ASIC system solution) in the most complicated situation, namely a 2D experiment with single shot measurement. However, some practical situations should be mentioned when a short reprogramming time can be desired. For example, for strong spin-echo signals the experiment can be performed at a higher temperature, i.e., at a higher repetition rate, and in turn in a shorter acquisition time. The overhead time, which can become comparable with the acquisition time, is not desired here.

Typical times required for the different parts of sequence processing are collected in Table 2. The times t_i, t_p, and t_1 correspond to the interpretation, the bit pattern creation, the loading of the data to the PFB, as shown in Fig. 5(b). The sum of these times, including a fixed waiting time for the boxcar integrator (50 μs), determines the minimum repetition time T_{rep}. According to Table 2, about half of the re-arm time is used for the calculation of the bit pattern. This is partially because the calculation also includes user-defined corrections for the pulse edge positions, as well as the control of the destination and the logic (positive/negative) of the pulses. In addition the state of the channel between the pulse events is controlled. For example the TWT gate has to cover the mw pulses, but when duration of the pulse sequence is longer than allowed TWT gate width, the gate signal will be splitted in two parts covering the corresponding mw pulses.

DISCUSSION

Tests and experiments have shown that the implemented system solution, "pulse language–real time controller–ASIC," allows one to build a high performance pulse programmer. The PPL simplifies the pulse sequence programming. The real time DSP controller is able to interpret the pulse program and to control the sequence flow in real time on a rate $>$ 1 kHz for experiments of any dimensions. For the simplest two-pulse experiment the repetition rate can be up to 2 kHz, as can be interpolated from Table 2. The short reprogramming time is reached by a proper design of the ASIC, a memory depth, which is small but sufficient for any reasonable number of pulses, and the possibility to repeat time intervals up to 2^{32}–2 times that allows for sufficiently long pulse sequences. In comparing the above-mentioned digital output generator (Chase Scientific Comp.) based on a general purpose AWG board has a similar, but more complicated organization of the pulse sequence processing. It includes memory segmentation, internal looping, and segment-to-segment jumping capabilities. As a consequence, the reprogramming time under conditions similar to those used in this work is ~2 ms (A. Astashkin, personal communication), which is about two times longer than for our programmer, but still well suitable for pulse EPR spectroscopy.

In perspective, a TMS320C64xx DSP could be used to further reduce the reprogramming time, and software optimization would reduce the time for interpretation and bit image creation. Moreover, an USB interface for DSP-host communication could be implemented to allow for effective data monitoring.

ACKNOWLEDGMENTS

The authors thank N. Felber for the assistance in the EPR ASIC design. This project has been supported by the Swiss National Science Foundation.

REFERENCES

1. Schweiger A, Jeschke G. 2001. Principles of pulse electron paramagnetic resonance. New York: Oxford University Press.
2. Shane JJ, Gromov I, Vega S, Goldfarb D. 1998. A versatile pulsed X-band ENDOR spectrometer. Rev Sci Instrum 69:3357–3364.
3. Quine RW, Harbridge JR, Eaton SS, Eaton GR. 1999. Design of a programmable timing unit. Rev Sci Instrum 70:4422–4432.
4. Fauth JM. 1988. Entwicklung von Pulssequenzen in der Elektronenspinecho-Spektroskopie, Ph.D. thesis. Swiss Federal Institute of Technology, Zurich.
5. Zhidkov VD, Borbat PP. 1991. Simple high precision pulse programmer for electron spin echo spectroscopy. ISMAR Workshop on Electron Spin Echo Spectroscopy, Novosibirsk, USSR, September 25–28, p 19A.
6. Wacker T. 1992. FID-detektiertes spektrales Lochbrennen in der Elektronen-Spin-Resonanz. Ph.D. thesis. Swiss Federal Institute of Technology, Zurich.
7. Gromov I, Shane J, Forrer J, Rakhmatoullin R, Rosenzwaig Y, Schweiger A. 2001. A Q-band pulse EPR/ENDOR spectrometer and the implementation of advanced one- and two-dimensional pulse EPR methodology. J Magn Reson 149:196–203.
8. Donnelly C, Stallman R. 1995. Bison. The YACC-compatible parser generator. Boston, MA: Free Software Foundation.
9. Gemperle C, Aebli G, Schweiger A, Ernst RR. 1990. Phase cycling in pulse EPR. J Magn Reson 88:241–256.
10. Almanac, Karlsruhe: Bruker Biospin; 2003.
11. Höfer P, Grupp A, Nebenfürh H, Mehring M. 1986. Hyperfine sublevel correlation (HYSCORE) spectroscopy: A 2D ESR investigation of the squaric acid radical. Chem Phys Lett 132:279–282.
12. AWG1000/1200 User Manual. 2004. http://chase2000.com/awg1000/awg1000.htm.